Plan For EAS Tagger

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1 Differential Intensity of Extensive Air Showers

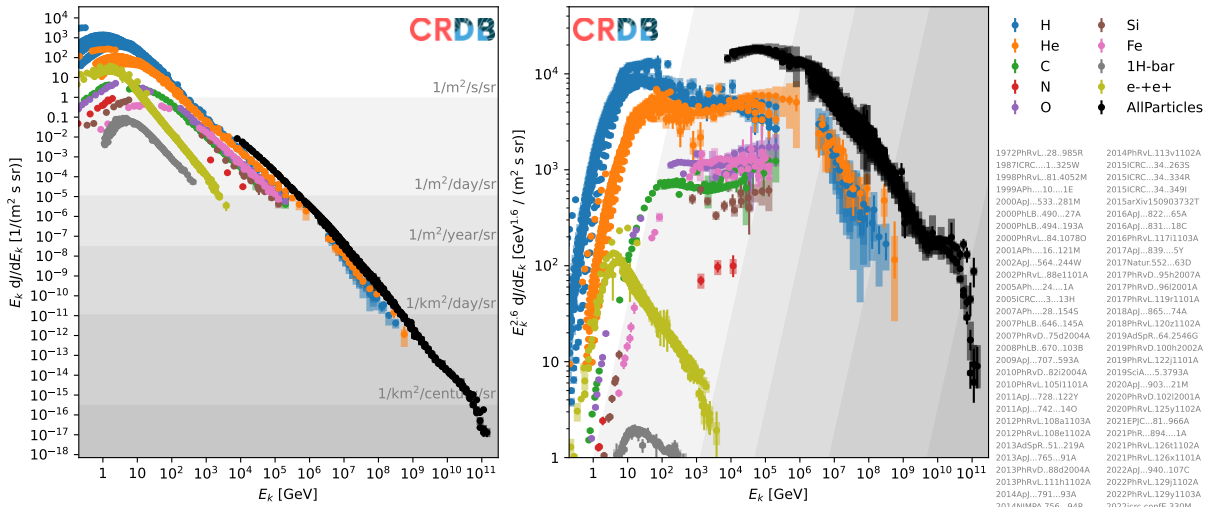


Figure 1: Incident differential intensity (dJ/dE), as presented by the Cosmic Ray Database (CRDB).

The differential[1] intensity dJ/dE of cosmic rays incident at the top of the atmosphere (TOA) is defined as

$$dJ/dE = \frac{dN}{dE\, dA\, dt\, d\Omega}$$

where E is the energy of the incident primary and t is time. The quantities $d\Omega$ and dA are the infinitesimal solid angle, relative to normal, and the infinitesimal area, both **at the TOA**. The differential intensity has units:

$$[dJ/dE] = \left[\frac{1}{\text{energy m}^2\text{ sr s}}\right]$$

where sr is a dimensionless reminder to integrate over solid angle. The incident differential intensity at the top of the atmosphere is independent of direction and follows a power law:

$$\frac{dJ}{dE} = \frac{\alpha}{\epsilon}\left(\frac{E}{\epsilon}\right)^{-\gamma} \tag{1}$$

where ϵ is a reference energy and $\gamma \sim 3$. The energy-integrated intensity above some energy E is:

$$J(E) \equiv \int_E^\infty \frac{dJ}{dE}\, dE = \frac{\alpha}{\gamma - 1}\left(\frac{E}{\epsilon}\right)^{1-\gamma} = \frac{E}{\gamma - 1}\frac{dJ}{dE} \tag{2}$$

[1]Take care that some sources use J for this quantity, but we are following the more explicit notation of e.g. the CRDB

which we can write as:

$$E\,\frac{dJ}{dE} = (\gamma - 1)\; J(E) \tag{3}$$

which shows that the y-axis in Fig. 1 is proportional to the integrated rate above, as indicated by the labeled gray bands. For example, there are:

$$24 \cdot 60 \cdot 60 \sim 10^5$$

seconds per day, so the once-per-day band appears at:

$$E\,\frac{dJ}{dE} = 10^{-5}\;\frac{1}{\mathrm{m}^2\;\mathrm{sr}\;\mathrm{s}}$$

Just take care, when interpreting these bands as integrated rates, that the factor of $\gamma - 1$ has been neglected, so the actual rates are about one half that indicated.

Equation 2 also implies that:

$$E\,\frac{dJ}{dE} = \alpha\,\left(\frac{E}{\epsilon}\right)^{1-\gamma} \tag{4}$$

which allows us to read off the parameters α and γ easily from Fig. 1. First note that:

$$E\;\left.\frac{dJ}{dE}\right|_{\epsilon} = \alpha$$

so we can read off α as the value of the y-axis for energy $E = \epsilon$. For example, taking $\epsilon = 10^9$ GeV we have

$$\alpha = 10^{-11}\;\frac{1}{\mathrm{m}^2\;\mathrm{sr}\;\mathrm{s}}$$

Next, taking the log of both sides of Equation 4 yields:

$$\ln\left(E\frac{dJ}{dE}\right) = \ln(\alpha) + (1-\gamma)\ln\left(\frac{E}{\epsilon}\right) \tag{5}$$

which shows that the power law is consistent with the linear relationship revealed in the log-log plot of Fig. 1. The y-axis drops by two orders of magnitude for every one order of magnitude increase of the x-axis, from which we read off:

$$1 - \gamma = -2 \implies \gamma = 3$$

To recap, the quantity

$$E\frac{dJ}{dE},$$

has several useful properties when plotted versus energy in a log-log scale, as in Fig. 1:

- The y-axis is proportional to integrated rate above E.
- The log-log plot is linear for a power-law.
- The parameter α can be estimated from the y-axis value for $E = \epsilon$.
- The parameter γ can be estimated from the slope.

The extent to which Fig. 1 is linear is a visual indication of the accuracy of the power law. We further improve the accuracy of the power-law model by selecting parameters near the region of interest.

2 Dimensionless Log-Energy Units

For computational work across many orders of magnitude in energy it is helpful to define and work with the dimensionless log-energy variable:

$$u \equiv \ln\left(\frac{E}{\epsilon}\right) \iff E = \epsilon\, e^u \tag{6}$$

from which it follows that:

$$du = \frac{dE}{E}$$

and

$$\frac{dJ}{du} = E\frac{dJ}{dE} \tag{7}$$

The power law of Equation 1 becomes:

$$\frac{dJ}{du} = \alpha \exp((1-\gamma)\, u) \tag{8}$$

and the linear relationship of Equation 5 becomes:

$$\ln \frac{dJ}{du} = \ln(\alpha) + (1-\gamma)u$$

Direct integration of Equation 8 gives:

$$J(u) \equiv \int_u^\infty \frac{dJ}{du}\, du = \frac{\alpha}{\gamma - 1} \exp((1-\gamma)\, u) = \frac{1}{\gamma - 1}\frac{dJ}{du} \tag{9}$$

which we could also obtain by substituting Equation 7 into Equation 2.

3 Shower Energy generation

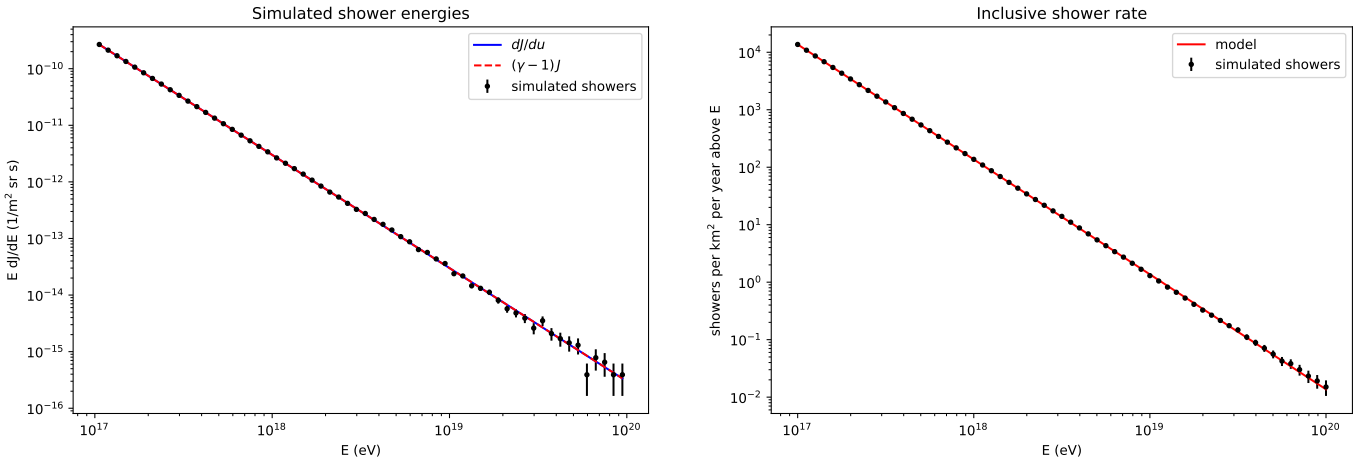
The power-law is a convenient model for both the analytic calculations that follow and for toy Monte Carlo shower generation which is used to validate our results. The power law is only approximate, or, equivalently, the parameters of the power law vary with shower energy. However, our results below include an estimate for a minimum detectable shower energy, and we expect that only a narrow energy region above this minimum will contribute significantly to our experimental rate. As long as the parameters for the power law are taken for the energy range appropriate for our experiment, we expect that our results will be accurate.

We simulate showers by drawing a random variable u, the dimensionless log-energy of the shower, following the distribution dJ/du. This is equivalent to drawing a primary energy E according to the power-law distribution dJ/dE, but is less susceptible to round-off errors. To avoid the divergence at $E = 0$, we set a minimum shower energy E_a and define:

$$a \equiv \ln(E_a/\epsilon)$$

The cumulative distribution function is:

$$P(u) = \frac{J(a) - J(u)}{J(a)}$$



(a) (b)

Figure 2: Simulated shower energy distribution (a) compared to two equivalent calculations of the power-law model, and (b) inclusive. Power law parameters are estimated at $E = 10^{18}$ eV $= 10^9$ GeV, where the simulation in (b) compares well to Fig. 1.

which is the probability that a shower with dimensionless log energy greater than a falls in the interval $[a, u]$. Equivalently,

$$P(u) = 1 - \exp\left((1-\gamma)\,(u-a)\right) \tag{10}$$

which is easily invertible. Therefore, to generate a random variable u, throw $0 \leq x < 1$ and calculate:

$$u = a - \frac{1}{\gamma - 1}\ln(1-x) \tag{11}$$

After generating N such showers, each should have a weight:

$$w = \frac{1}{N}\,J(a) = \frac{1}{N}\,\frac{\alpha}{\gamma-1}\,\exp\left((1-\gamma)\,a\,\right)$$

which accounts for the original assumption that the shower has $u > a$.

4 Incident Cosmic Ray Flux

Next we calcualte the flux of incident cosmic ray particles when projected (without any attenuation or scattering) to the surface of the earth. We approximate the earth as an infinite plane (appropriate for our thin atmosphere, as we shall see) and calculate the flat-surface differential flux as:

$$F = \int J\cos\theta d\Omega = 2\pi J\int_0^{\pi/2}\cos\theta\,\sin\theta\,d\theta$$

where we integrate only over the upper hemisphere, have included a factor of $\cos\theta$ for a flat horizontal surface, and used the fact that J is independent of direction. Computing the integral we find that:

$$F = C\pi J \tag{12}$$

or equivently:

$$\frac{dF}{dE} = C\pi\frac{dJ}{dE} \tag{13}$$

where $C = 1$ for the flat-surface approximation made here. The effect of the Earth's curvature is considered in the Appendix, where it shown that $C = 0.92$.

5 Local Muon Density from Extensive Air Showers

The number of muons N produced by a shower of energy E follows a power law:

$$N(E) = \beta \left(\frac{E}{\epsilon}\right)^{\eta} \tag{14}$$

Suppose that these N muons have lateral density profile given by:

$$\Phi(r) = \Phi(0) \exp(-r/\lambda)$$

where λ is the length scale of the muon lateral profile. The condition:

$$\int_0^\infty \Phi(r)\, 2\pi r\, dr = N$$

implies that:

$$\Phi(0) = \frac{N(E)}{2\pi\lambda^2} \tag{15}$$

and

$$\Phi(r) = \frac{N(E)}{2\pi\lambda^2}\ \exp(-r/\lambda) \tag{16}$$

This is invertible to find an equation for the radius at which the density falls to Φ:

$$r(\Phi) = -\lambda \ln\left(\frac{2\pi\lambda^2\Phi}{N(E)}\right) \tag{17}$$

6 Local Muon Flux from Extensive Air Showers

A critical consideration for the feasibility of studying extensive air showers in the lab, using a local network of cell phones, is the rate of extensive air showers which cause the muon flux in the lab to reach or exceed a level detectable by the phone network. A typical phone has an area times efficiency of

$$A_{\text{eff}} = 2 \times 10^{-5}\ \text{m}^2$$

We define a detectable muon flux Φ_D as the density at which a network of N_{P} phones is expected to register at least one hit on average. So for $N_{\text{P}} = 1000$, we have:

$$\Phi_{\text{D}} = \frac{1}{A_{\text{eff}}\ N_{\text{P}}} = 50\ \text{muons/m}^2 \tag{18}$$

We saw in the preceding section that the incident flux of cosmic rays follows a power law, and the differential intensity is given by

$$\frac{dJ}{dE} = \frac{\alpha}{\epsilon}\left(\frac{E}{\epsilon}\right)^{-\gamma}$$

and the angle-integrated differential flux:

$$\frac{dF}{dE} = 0.92\ \pi\ \frac{dJ}{dE}$$

where we can set 0.92 to one if we want to approximate the earth as an infinite plane.

which means the area $A_{\rm D}$ over which there is a detectable muon flux is given by:

$$A_{\rm D}(E) = \pi\lambda^2 \left(\ln \frac{2\pi\lambda^2\Phi_{\rm D}}{N(E)}\right)^2$$

which is only valid ($A_{\rm D} > 0$) when:

$$\Phi(0) = \frac{N(E)}{2\pi\lambda^2} > \Phi_D$$

which sets a minimum energy, as we shall see. Using the power law, note that:

$$\frac{N(E)}{\Phi_{\rm D}\, 2\pi\lambda^2} = \frac{\beta}{\Phi_{\rm D}\, 2\pi\lambda^2}\left(\frac{E}{\epsilon}\right)^{\eta} = \left(\frac{E}{\kappa\epsilon}\right)^{\eta}$$

where

$$\kappa = \left(\frac{\Phi_{\rm D} 2\pi\lambda^2}{\beta}\right)^{1/\eta} \tag{19}$$

and so:

$$A_{\rm D}(E) = \pi\lambda^2\eta^2 \left(\ln \frac{E}{\kappa\epsilon}\right)^2 \tag{20}$$

which is only valid for $E > \kappa\epsilon$ (which you can verify is equivalent to requiring that $\Phi(0) > \Phi_{\rm D}$).

We are interested in the quantity:

$$\frac{dN}{dEdt} = \frac{dF}{dE}\, A_{\rm D}(E) = (0.92)\pi\frac{dJ}{dE}A_{\rm D}(E) \tag{21}$$

Plugging in the results below we obtain:

$$\frac{dN}{dE\, dt} = (0.92)\frac{\alpha}{\epsilon}\,(\pi\lambda\eta)^2 \left(\frac{E}{\epsilon}\right)^{-\gamma}\left(\ln \frac{E}{\kappa\epsilon}\right)^2 \tag{22}$$

which we can integrate from $E = \kappa\epsilon$ to ∞ and obtain our main result:

$$\frac{dN}{dt} = (0.92)\alpha\,(\pi\lambda\eta)^2\kappa^{1-\gamma}\,\frac{2}{(\gamma-1)^3} \tag{23}$$

7 Projections

- The cosmic ray background is 170 muons per second per m^2. If phones integrate 1 s, that amounts to 170 muons per m^2 of muon flux, which any signal must overcome.
- We could try to directly reduce this background, e.g. by siting the pilot study underground.
- We could improve the timing resolution of the phones. At 1/10 s the background density becomes 17 muons per m^2, and at 1/100 s, 1.7 muons per m^2.
- If our tagger is set to trigger at muon flux higher than the background flux, we can reliably get a signal that exceeds background. But our sample size is limited by the rate at which EAS cause showers. For example (with $\lambda = 200m$)

muons / m^2	showers per year
340	0.02
100	0.4
50	2.0
20	14
10	63
5	282

- We could add sites or years to improve these statistics, but rejecting background seems more promising to me.
- Almost certainly we can get a factor of 1/10 reduction, with limited additional deadtime, by using an external trigger. Perhaps even we can do this on the phone, with a better designed pipeline.
- Getting a factor of 1/100th might be possible with external trigger, but deadtime is likely becoming significant (50% assuming readout is 10 ms).
- 1/10 time resolution plus overburden 1/10 gets us to 1.7 muons / m^2, and 282 events per year well exceeding this...

A Effect of Earth's Curvature

Our earth has atmosphere of height h and radius R. An incident cosmic ray primary has some impact parameter b, which is the same whethere measured at TOA or on the ground. At TOA we have incident angle θ_h and:

$$\sin\theta_h = \frac{b}{R+h}$$

while at ground we have incident angle θ_g and:

$$\sin\theta_g = \frac{b}{R}$$

so that:

$$(R+h)\sin\theta_h = R\sin\theta_g \tag{24}$$

The Jacobian:

$$\frac{d\Omega_h}{d\Omega_g} = \left(\frac{R}{R+h}\right)^2 \frac{\cos\theta_g}{\sqrt{1-\left(\frac{R}{R+h}\right)^2\sin^2\theta_g}}$$

and the resulting incident flat-surface differential flux (now accounting for earth's shape) is:

$$F = 0.92\,\pi\,J$$

for TOA at $h = 100$ km and earth radius $R = 6371$ km.

B Analysis tables

Written by the analysis notebooks (analysis/tex/).

		value	unit
ϵ	reference energy	10^{18}	eV
α	intensity at ϵ, $dJ/dE = (\alpha/\epsilon)(E/\epsilon)^{-\gamma}$	3×10^{-12}	$\mathrm{m^{-2}\ sr^{-1}\ s^{-1}}$
γ	spectral index	3	
β	muons at ϵ, $N = \beta(E/\epsilon)^{\eta}$ (CORSIKA fit)	7.28×10^{6}	
η	muon number index (CORSIKA fit)	0.936	
λ	lateral length of the exponential profile	100	m
C	Earth's curvature	0.92	
Φ_D	detectable muon density	$50, 20$	$\mathrm{m^{-2}}$
Φ_μ	background muon flux	170	$\mathrm{m^{-2}\ s^{-1}}$
N_P	phones	1000	
A_P	area × efficiency per phone	2×10^{-5}	$\mathrm{m^2}$
Δt_P	phone window (fixed)	0.1	s
N_T	taggers	10	
A_T	area × efficiency per tagger	0.014	$\mathrm{m^2}$
Δt_T	tagger coincidence window (sliding)	10^{-6}	s

Table 1: Model parameters, from model.yaml and nmuons.yaml (full_analytic).

$\Phi_D = 50\ \mathrm{m^{-2}}$, one year	
showers per year, every one at Φ_D	11.3
taggers (10 of 0.014 $\mathrm{m^2}$, 1 μs), at least 3 fired	
shower tags per year	10.7
combinatoric tags per year	0.15
phones at the tags (1000 of 0.2 $\mathrm{cm^2}$, 0.1 s), fired per year	
signal	10.7
background	3.70
significance	4.0 σ

Table 2: TODO. Tagging at threshold: the operating point, phones read out at the tags, and the significance of the summed excess (tagging_analytic).

C Analysis figures

Stubs: each figure from the analysis notebooks not yet in the text.

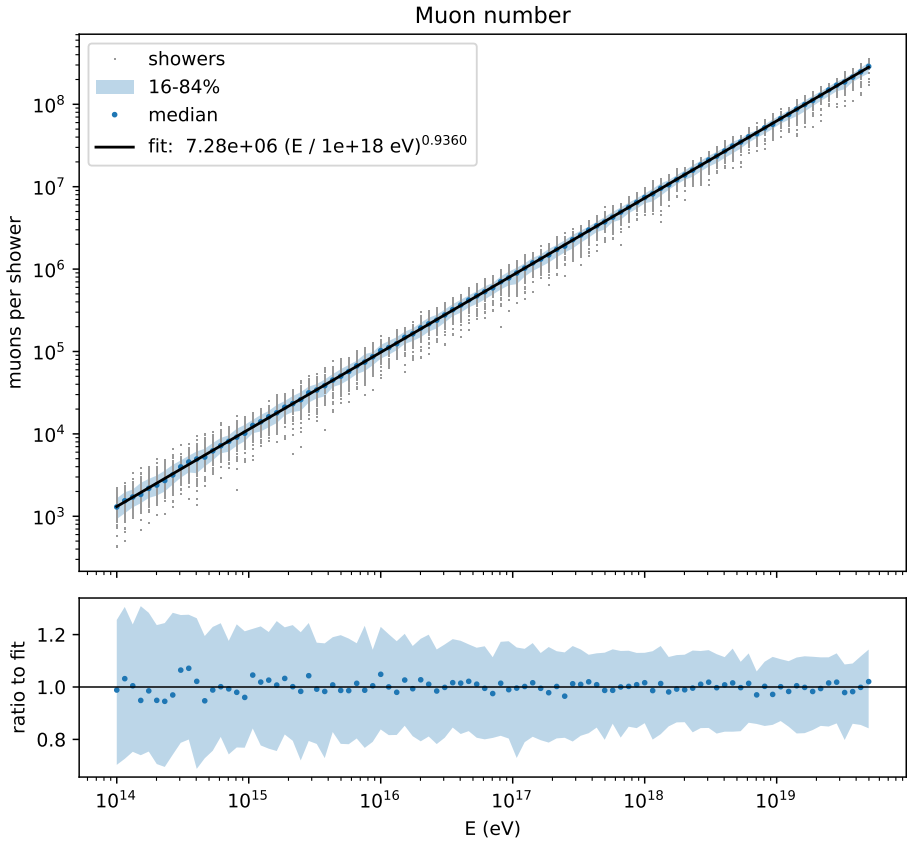


Figure 3: TODO. Muon number per shower from CORSIKA (aaa_muons).

D Introduction

- Using rates from CRDB.
- Using Whitesons formula for the number of muons:

$$N_\mu = 0.028(E/1\text{ GeV})^{0.93}$$

  Note that for $E = 10^{19}$ eV, this formular predicts 5×10^7 in good agreement with LHS of Fig. 16

- Muons have typical lateral extent of λ greater than Moliere radius. Assuming:

$$\rho(r) = \frac{N}{2\pi\lambda^2}\ \exp(-x/\lambda)$$

  then $\lambda = 220$ puts $\rho(1\text{km}) = 2/\text{m}^2$ as in figure.

- Aim for $E \geq 10^{18}$ eV.

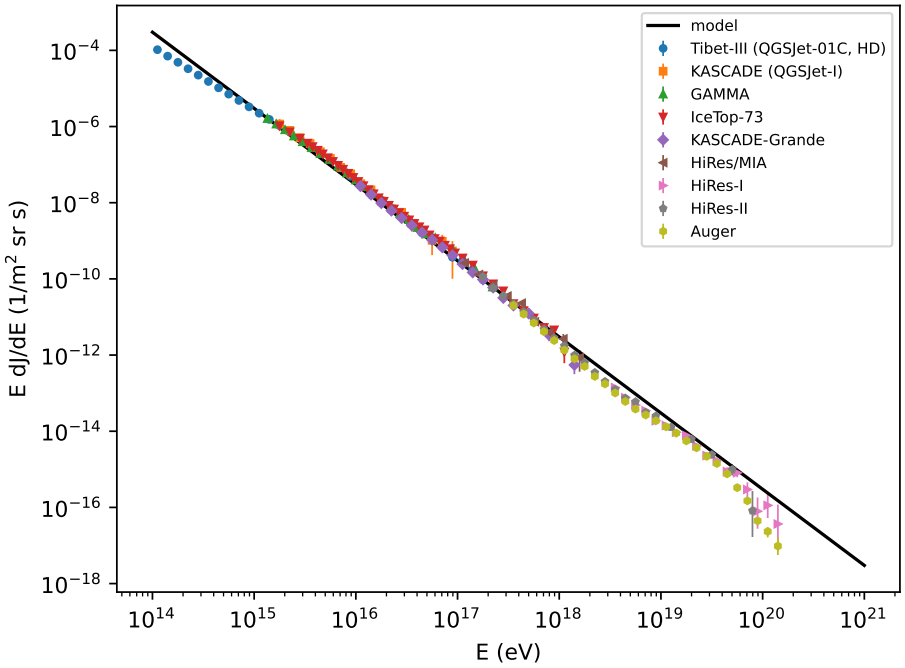


Figure 4: TODO. Spectrum in the analytic model against the data (full_analytic).

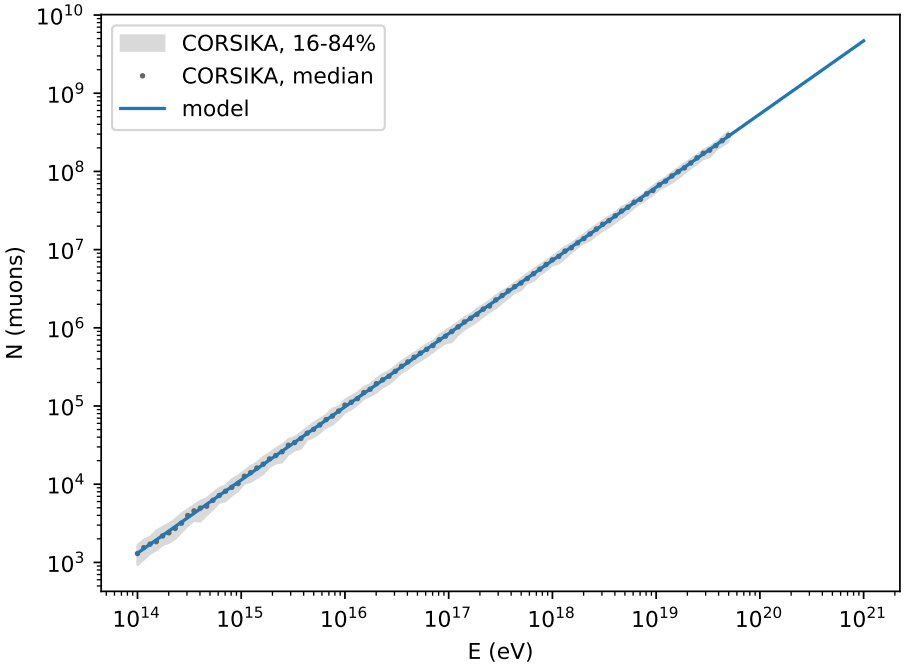


Figure 5: TODO. Muon number in the analytic model against CORSIKA (full_analytic).

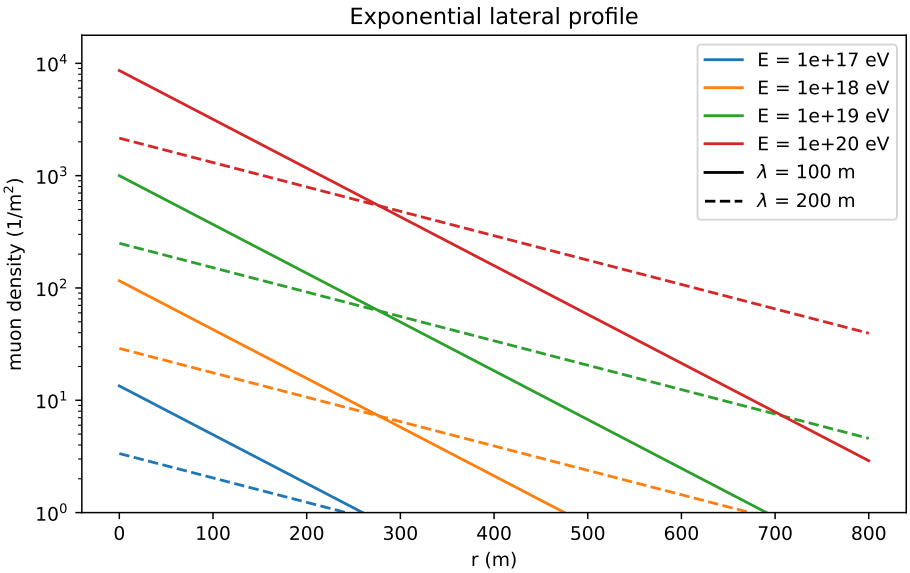


Figure 6: TODO. Exponential lateral profile (full_analytic).

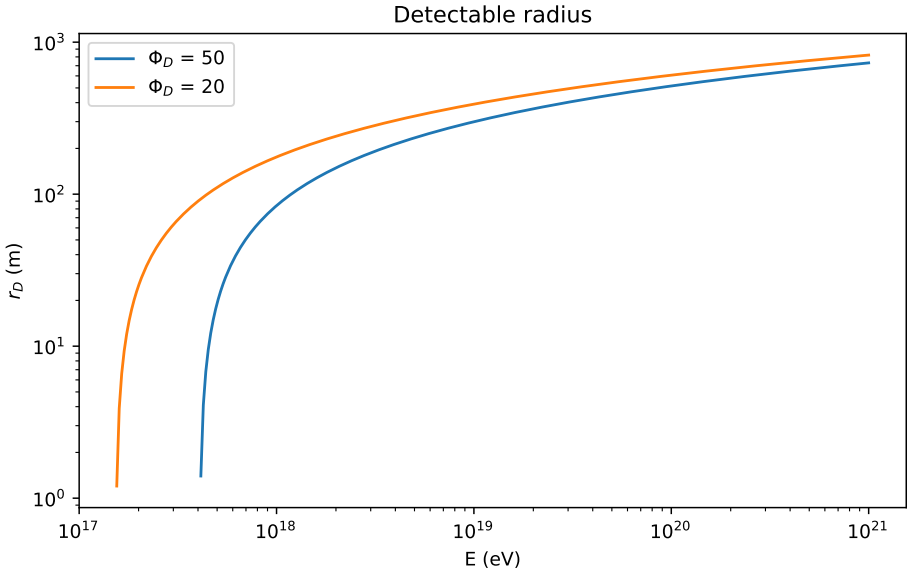


Figure 7: TODO. Detectable radius (full_analytic).

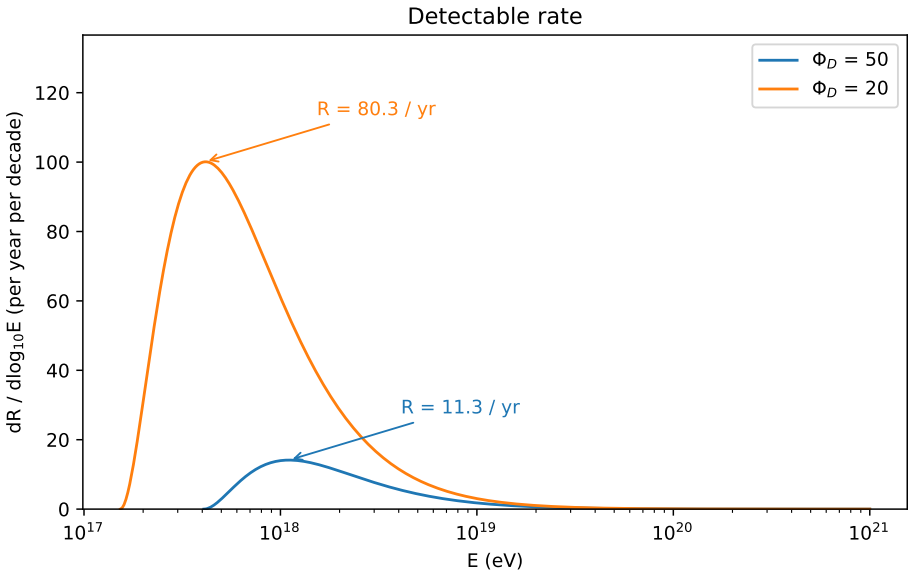


Figure 8: TODO. Detectable rate in the analytic model (full_analytic).

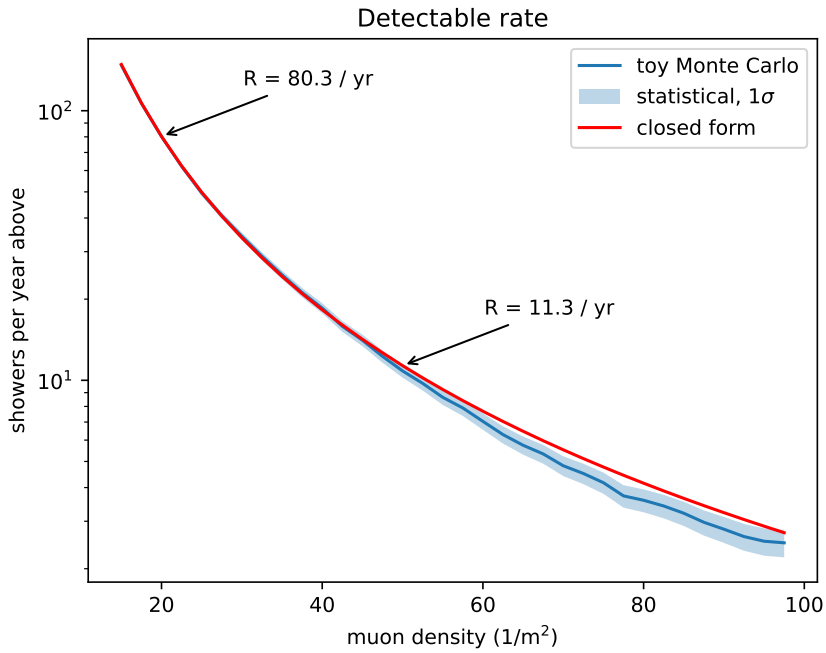


Figure 9: TODO. Detectable rate from the toy Monte Carlo against the closed form (shower_toy).

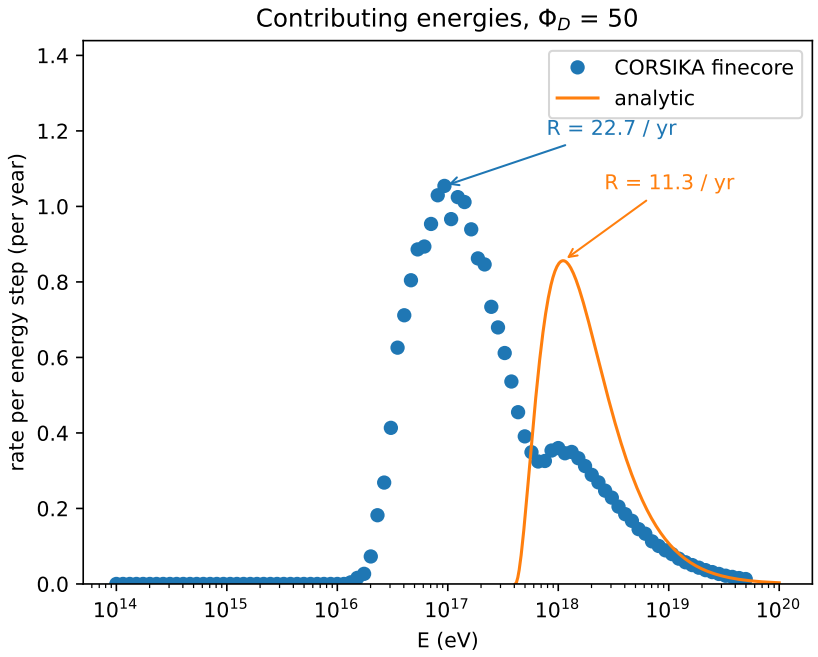


Figure 10: TODO. Contributing shower energies at $\Phi_D = 50$, CORSIKA against the analytic model (corsika_rate).

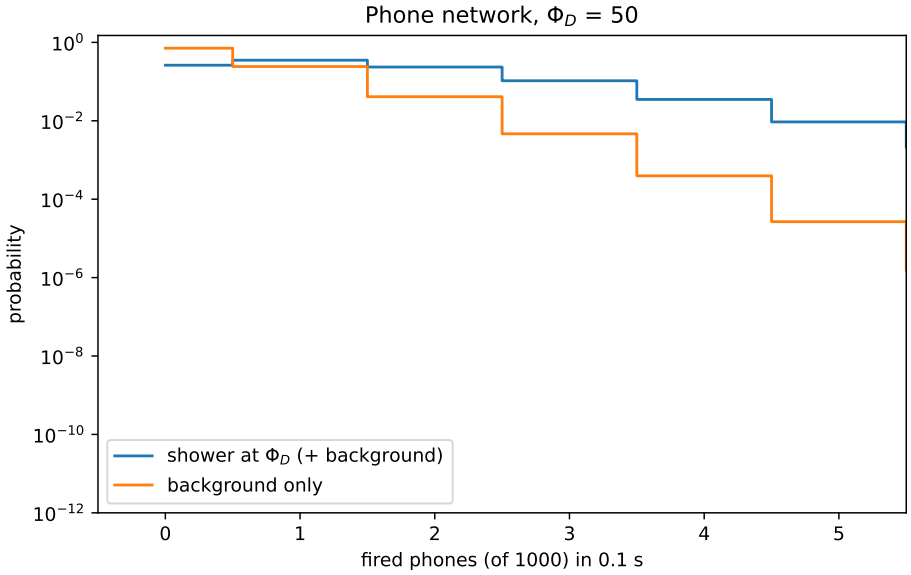


Figure 11: TODO. Fired phones in one window, a shower at Φ_D against background alone (tagging_analytic).

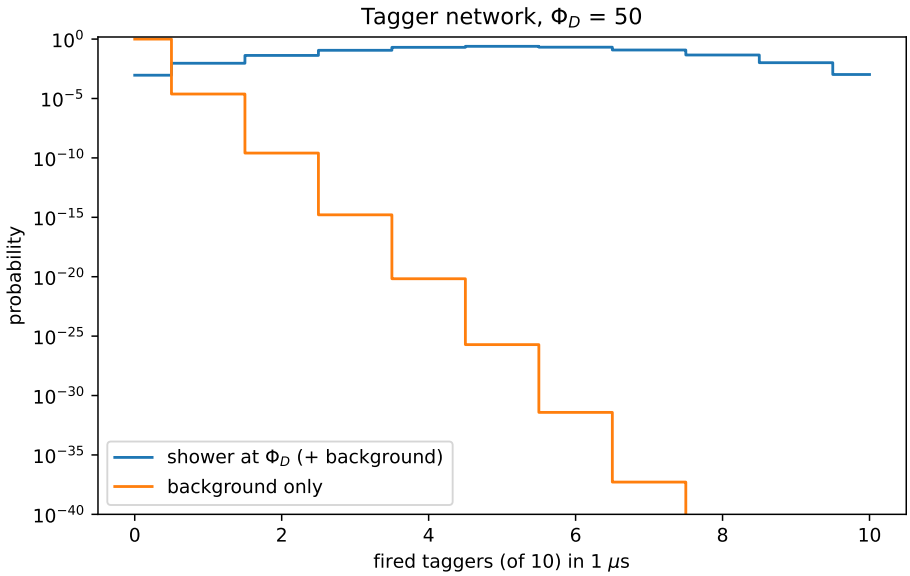


Figure 12: TODO. Fired taggers in one window, a shower at Φ_D against background alone (tagging_analytic).

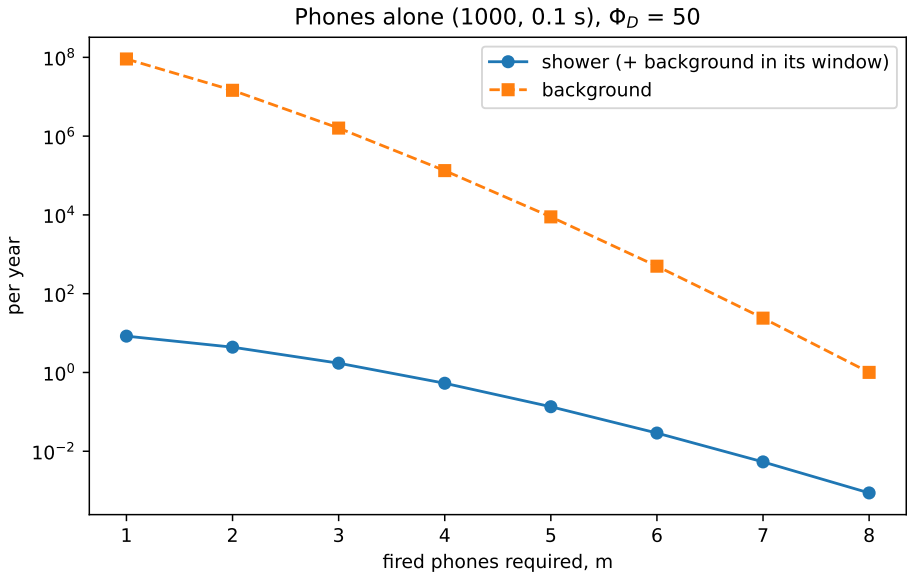


Figure 13: TODO. Phones alone, untagged: counts per year against the fired phones required (tagging_analytic).

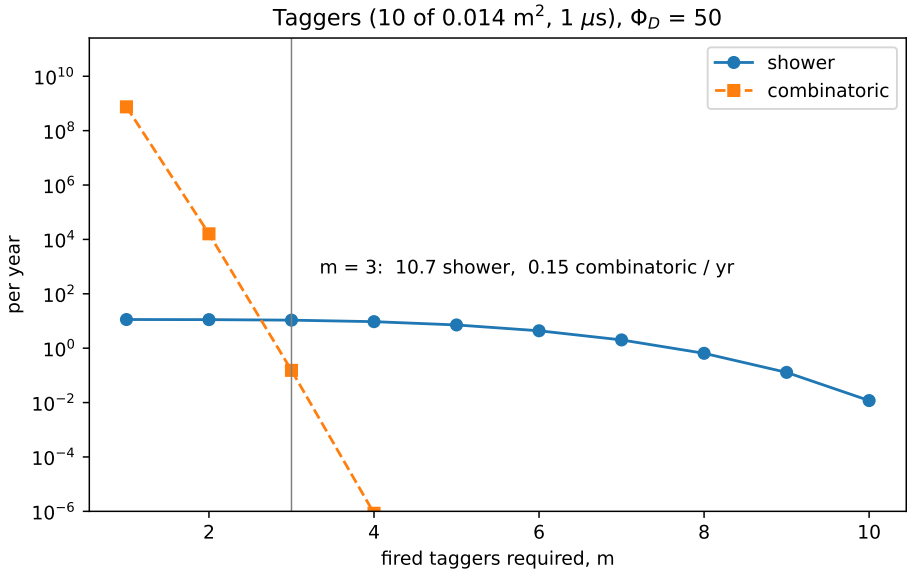


Figure 14: TODO. Taggers: shower and combinatoric tags per year against the fired taggers required; the operating point marked (tagging_analytic).

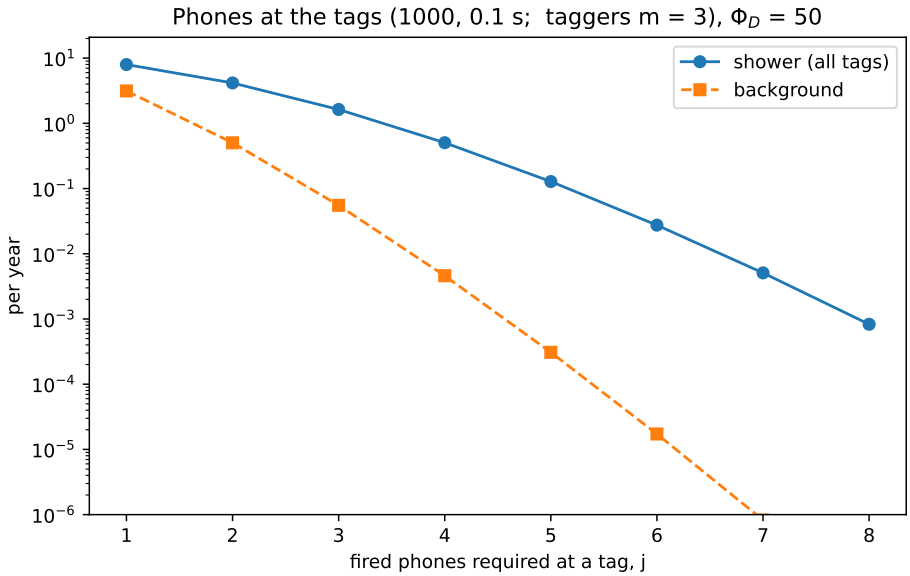


Figure 15: TODO. Phones read out at the tags: counts per year against the fired phones required (tagging_analytic).

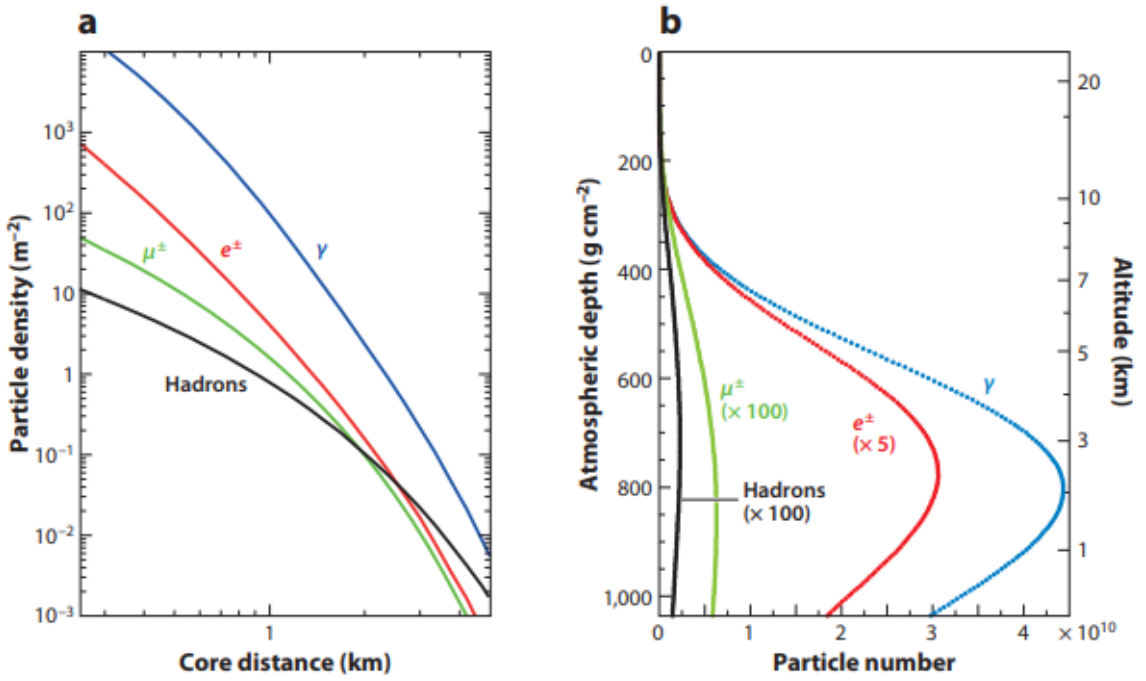


Figure 2

Average (*a*) lateral and (*b*) longitudinal shower profiles for vertical, proton-induced showers at 10^{19} eV. The lateral distribution of the particles at ground is calculated for 870 g cm^{-2}, the depth of the Pierre Auger Observatory. The energy thresholds of the simulation were 0.25 MeV for γ and $e^{\pm}$ and 0.1 GeV for muons and hadrons.

Figure 16: Extensive air showers, from Chapter 16 of "Gaissner Cosmic Rays and Particle Physics"

- Expected rate is:
$$R = \frac{1}{\mathrm{km}^2\,\mathrm{day}\,\mathrm{sr}}\,\pi(\pi\lambda^2) = 0.44\,/\,\mathrm{day}$$
- Total muons:
$$N_\mu = 6.6 \times 10^6$$
- And density at center:
$$\rho(0) = \frac{N_\mu}{2\pi\lambda^2} = 21/\ \mathrm{m}^2$$
- $A\epsilon = 2 \times 10^{-5}$, for 1000 phones: 0.02 m^2
- Muons hits in phone per shower: 0.42 per shower.
- Expect a phone hit coincident with tagger more than once per week.
- Taggers of size $0.1m^2$ should be well suited.